Lambda1/100/1000 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 17 complete ICs for all stages within ANN10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda / stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
1 ANN10	**1 / soft10k**	0.204	0.799	4.12	0.99	0.613	2.46	0.265	2.84	2.51	0	17/18
1 ANN10	**1 / long**	0.172	0.683	3.8	0.963	0.524	1.81	0.203	6.25	27.4	0	18/18
1 ANN10	**100 / soft10k**	0.658	0.758	1	1	1.52	26.9	1	1.06	59.3	0	18/18
1 ANN10	**100 / long**	0.664	0.891	1	1	0.971	32.5	1	13.6	6.02e+03	0	18/18
1 ANN10	**1000 / soft10k**	0.681	1.36	1	78.4	4.24	4.14	1	0.999	653	0	18/18
1 ANN10	**1000 / long**	0.645	1.08	1	1	2.89	24.8	1	1	14.3	0	18/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 4, 5, 7, 18, 32, 39, 44, 45, 53, 64, 68, 69, 73, 74, 76, 78, 88. Team-specific cohorts differ; do not rank architectures using different IC sets.

Lambda1/100/1000 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 1 complete ICs for all stages within P-only DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda / stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
2 P-only DeepONet10	**1 / soft10k**	0.109	0.2	0.888	0.462	1.24	7.23	0.253	0.999	5.42	0	18/18
2 P-only DeepONet10	**1 / long**	0.266	0.11	2.1	30.9	2.23	0.979	0.182	2.31e+03	1.55e+03	0	18/18
2 P-only DeepONet10	**100 / soft10k**	0.593	0.718	1	1	9.71	28.3	1	1.02	3.14	0	6/18
2 P-only DeepONet10	**100 / long**	0.602	0.66	1	1	7.39	30.2	1	0.997	1.02	0	1/18
2 P-only DeepONet10	**1000 / soft10k**	0.522	0.942	1	1	12.3	1	1	4.93e+04	1	0	18/18
2 P-only DeepONet10	**1000 / long**	0.522	0.952	1	1	13	1	1	4.77e+04	1	0	18/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 4. Team-specific cohorts differ; do not rank architectures using different IC sets.

Lambda1/100/1000 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 13 complete ICs for all stages within ANN10 + P-only DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda / stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
3 ANN10 + P-only DeepONet10	**1 / soft10k**	0.0354	0.215	0.679	5.83	0.598	0.988	0.0743	8.18	1.83	0	18/18
3 ANN10 + P-only DeepONet10	**1 / long**	0.0716	0.172	0.357	0.525	0.449	0.865	0.0749	3.57	1.05	0	18/18
3 ANN10 + P-only DeepONet10	**100 / soft10k**	0.641	1.04	1	1	2.72	26.3	1	1.17	43.5	0	18/18
3 ANN10 + P-only DeepONet10	**100 / long**	0.416	0.919	1	1.44	1.44	19.8	1	8.33	1.52e+03	0	13/18
3 ANN10 + P-only DeepONet10	**1000 / soft10k**	0.544	2.15	1	1	8.81	1	1	0.998	2.1e+03	0	18/18
3 ANN10 + P-only DeepONet10	**1000 / long**	0.581	1.97	1	0.999	7.56	10.6	1	0.999	19.5	0	18/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 4, 18, 32, 39, 44, 45, 53, 57, 68, 69, 73, 76, 78. Team-specific cohorts differ; do not rank architectures using different IC sets.

Lambda1/100/1000 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 1 complete ICs for all stages within T/P DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda / stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
4 T/P DeepONet10	**1 / soft10k**	1.29	0.927	9.81e+07	2.14e+08	0.485	6.39e+08	1.87e+06	188	808	0	9/18
4 T/P DeepONet10	**1 / long**	1.15	0.968	1.13e+08	4.38e+08	0.557	6.62e+09	1.78e+06	354	4.30e+04	0	14/18
4 T/P DeepONet10	**100 / soft10k**	0.194	0.628	4.32e+07	2.41e+08	0.616	3.99e+10	1.49e+05	1	1	0	18/18
4 T/P DeepONet10	**100 / long**	0.113	0.744	7.35e+04	3.8	0.715	4.46e+10	2.45e+05	1	1	0	3/18
4 T/P DeepONet10	**1000 / soft10k**	1.08e-04	9.71e-04	1	1	2.03e-03	1	1	2.14e+03	1	0	18/18
4 T/P DeepONet10	**1000 / long**	2.47e-04	3.35e-04	1	0.996	3.88e-04	1	0.998	81	866	0	18/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 69. Team-specific cohorts differ; do not rank architectures using different IC sets.

Lambda1/100/1000 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 18 complete ICs for all stages within ANN10 + T/P DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda / stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
5 ANN10 + T/P DeepONet10	**1 / soft10k**	0.0803	0.41	0.932	0.819	0.535	0.822	0.103	0.938	1.16	0	18/18
5 ANN10 + T/P DeepONet10	**1 / long**	0.0902	0.189	0.888	0.588	0.4	0.518	0.0959	0.905	0.992	0	18/18
5 ANN10 + T/P DeepONet10	**100 / soft10k**	0.332	0.973	1.42	11.5	2.46	1	0.561	324	2.03e+03	0	18/18
5 ANN10 + T/P DeepONet10	**100 / long**	0.204	0.783	1	3.18	0.925	2.98	0.442	303	1.32e+04	0	18/18
5 ANN10 + T/P DeepONet10	**1000 / soft10k**	0.542	1.64	1	1	5	1	1	0.998	761	0	18/18
5 ANN10 + T/P DeepONet10	**1000 / long**	0.544	1.64	1	1	5.01	1	1	0.999	8.34	0	18/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88. Team-specific cohorts differ; do not rank architectures using different IC sets.

Lambda1/100/1000 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 4 complete ICs for all stages within T/P/ϕ DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda / stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
6 T/P/ϕ DeepONet10	**1 / soft10k**	1.57	0.908	4.65e+05	1.62e+07	0.849	2.81e+05	5.94e+05	18.8	0.959	0	11/18
6 T/P/ϕ DeepONet10	**1 / long**	0.699	0.493	9.23e+06	3.61e+07	0.458	6.41e+08	5.04e+05	99.1	10.8	0	13/18
6 T/P/ϕ DeepONet10	**100 / soft10k**	0.239	0.59	1.04e+08	3.20e+08	0.812	1.17e+10	1.55e+05	1	1	0	16/18
6 T/P/ϕ DeepONet10	**100 / long**	0.121	0.191	3.06	1.24e+03	0.483	2.24e+07	0.986	13.7	5.37e+04	0	4/18
6 T/P/ϕ DeepONet10	**1000 / soft10k**	1.56e-03	2.16e-03	1	1	2.48e-03	1	1	70.5	1.14	0	18/18
6 T/P/ϕ DeepONet10	**1000 / long**	2.93e-05	2.36e-05	1	1.02	3.82e-05	1	1	1.04	6.04	0	18/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 4, 7, 69, 78. Team-specific cohorts differ; do not rank architectures using different IC sets.

Lambda1/100/1000 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 17 complete ICs for all stages within ANN10 + T/P/ϕ DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda / stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
7 ANN10 + T/P/ϕ DeepONet10	**1 / soft10k**	0.395	0.881	2.89	3.51	0.521	8.19	0.507	1.22	1.15	0	17/18
7 ANN10 + T/P/ϕ DeepONet10	**1 / long**	0.209	0.347	1.2	1.07	0.549	1.15	0.232	1.29	1.27	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	**100 / soft10k**	0.349	0.865	1.55	11.3	3.53	1	0.587	358	2.88e+03	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	**100 / long**	0.669	0.765	2.77	4.56	0.971	19.3	1	215	2.04e+04	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	**1000 / soft10k**	0.545	1.47	1	1	5.06	1	1	0.998	752	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	**1000 / long**	0.545	1.48	1	0.999	5.09	1	1	1	3.8	0	18/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 4, 5, 7, 18, 32, 39, 44, 45, 53, 64, 68, 69, 73, 74, 76, 78, 88. Team-specific cohorts differ; do not rank architectures using different IC sets.